Network Scanning Lab Report (Lab 10)

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1 Introduction

This report documents the Network Scanning Lab (Lab 10), which focuses on using tools like Nmap, Wireshark, and Metasploit to perform network scanning, port scanning, service detection, and vulnerability assessment. The lab covers various scanning techniques, evasion methods, and scripting with the Nmap Scripting Engine (NSE). The objectives include identifying live hosts, open ports, services, operating systems, and potential vulnerabilities in a controlled lab environment.

The lab was conducted on a Linux system (Kali Linux) with a target network consisting of virtual machines (e.g., IP addresses 172.16.4.2421 and 172.16.4.2422). This report includes step-by-step explanations, screenshots of key outputs, and personal observations for each task.

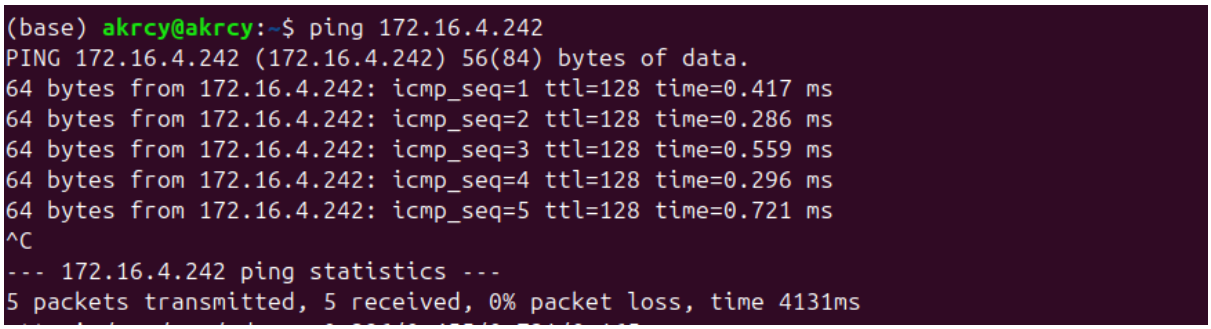
2 Network Discovery

2.1 Ping Scan

Command: `ping 172.16.4.242`

Explanation: The ping command sends ICMP echo requests to check if a host is reachable. A successful response indicates the host is up. The TTL value (e.g., 64) can hint at the operating system (Linux in this case).

Screenshot: **Observation**: The host responded with a TTL of 128, suggesting WIndows-



```
(base) akrcy@akrcy:~$ ping 172.16.4.242
PING 172.16.4.242 (172.16.4.242) 56(84) bytes of data.
64 bytes from 172.16.4.242: icmp_seq=1 ttl=128 time=0.417 ms
64 bytes from 172.16.4.242: icmp_seq=2 ttl=128 time=0.286 ms
64 bytes from 172.16.4.242: icmp_seq=3 ttl=128 time=0.559 ms
64 bytes from 172.16.4.242: icmp_seq=4 ttl=128 time=0.296 ms
64 bytes from 172.16.4.242: icmp_seq=5 ttl=128 time=0.721 ms
^C
--- 172.16.4.242 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4131ms
```

Figure 1: Enter Caption

based system. The response time was around 4131 ms, indicating a local network.

2.2 Nmap Ping Scan

Command: `sudo nmap -sn 172.16.4.242`

Explanation: Nmap's ping scan (-sn) confirms host availability without scanning ports, making it faster. It also provides the MAC address and vendor information.

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -sn 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:29 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00020s latency).
MAC Address: 70:B5:E8:58:22:8E (Dell)
Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
```

Figure 2: Enter Caption

Observation: The host was up with a latency of 0.00020s and mac Adrdress show it is a Dell device.

2.3 ARP Scan

Command: `arp -e`

Explanation: The ARP command lists devices on the local network, showing IP and MAC addresses. This helps map the network topology.

Screenshot:

```
(base) akrcy@akrcy:~$ arp -e
Address                  HWtype  HWaddress           Flags Mask            Iface
169.254.39.184           ether   10:e7:c6:fe:a4:b1   C                     enp1s0
_gateway                 ether   74:78:a6:a7:52:f7   C                     wlp2s0
172.16.4.135                     (incomplete)                              enp1s0
_gateway                 ether   2c:d0:2d:4c:11:58   C                     enp1s0
172.16.4.242             ether   70:b5:e8:58:22:8e   C                     enp1s0
172.16.4.182             ether   00:23:81:2e:90:b2   C                     enp1s0
(base) akrcy@akrcy:~$
```

Figure 3: Enter Caption

Observation: Multiple devices were listed, including the target 172.16.4.242, confirming its presence on the subnet.

3 Port Scanning

3.1 Single Port Scan

Command: `sudo nmap -p 80 172.16.4.242`

Explanation: This scans port 80 (HTTP) to check its state (open, closed, or filtered). An open port indicates a running service.

Screenshot:

Observation: The host at 172.16.4.242 is active, but port 80 is filtered, likely due to a firewall or packet filtering that blocks external scans.

```
(base) akrcy@akrcy:~$ sudo nmap -p 80 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:33 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00015s latency).

PORT   STATE    SERVICE
80/tcp filtered http
MAC Address: 70:B5:E8:58:22:8E (Dell)

Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
(base) akrcy@akrcy:~$
```

Figure 4: Enter Caption

3.2 Multiple Port Scan

Command: `sudo nmap -p 21,80,22 172.16.4.2422`

Explanation: Scans multiple ports (FTP, HTTP, SSH) in one command, identifying which services are active.

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -p 21,80,22 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:35 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00060s latency).

PORT   STATE    SERVICE
21/tcp filtered ftp
22/tcp filtered ssh
80/tcp filtered http
MAC Address: 70:B5:E8:58:22:8E (Dell)
```

Figure 5: Enter Caption

Observation: The host at 172.16.4.242 is active, but port 80,22,21 are filtered.

3.3 All Ports Scan

Command: `sudo nmap -p- 172.16.4.242`

Explanation: Scans all 65,535 TCP ports to ensure no services are missed. This is thorough but time-consuming.

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -p- 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:37 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00055s latency).
Not shown: 65528 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
5040/tcp  open  unknown
5357/tcp  open  wsdapi
7070/tcp  open  realserver
49668/tcp open  unknown
MAC Address: 70:B5:E8:58:22:8E (Dell)

Nmap done: 1 IP address (1 host up) scanned in 104.46 seconds
```

Figure 6: Enter Caption

Observation:Ports 135, 139,445 etc were open,

3.4 UDP Scan

Command: `sudo nmap -sU 172.16.4.242`

Explanation: Scans UDP ports, which are less common but critical for services like DNS (port 53).

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -sU 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:41 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00024s latency).
Not shown: 999 open|filtered udp ports (no-response)
PORT    STATE SERVICE
137/udp open  netbios-ns
MAC Address: 70:B5:E8:58:22:8E (Dell)

Nmap done: 1 IP address (1 host up) scanned in 13.66 seconds
```

Figure 7: Enter Caption

Observation: UDP ports 135 is open, indicating potential DNS and DHCP services.

4 Service and OS Detection

4.1 Version Scan

Command: `sudo nmap -sV 172.16.4.242`

Explanation: Identifies service versions (e.g., Apache 2.4.58) running on open ports, aiding vulnerability assessment.

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -sV 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:42 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00053s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT     STATE SERVICE         VERSION
135/tcp  open  msrpc           Microsoft Windows RPC
139/tcp  open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp  open  microsoft-ds?
5357/tcp open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
7070/tcp open  ssl/realserver?
MAC Address: 70:B5:E8:58:22:8E (Dell)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/sub
Nmap done: 1 IP address (1 host up) scanned in 16.25 seconds
```

Figure 8: Enter Caption

Observation: The target host (172.16.4.242) is a Windows machine, likely running services related to file sharing and possibly media streaming.

4.2 OS Detection

Command: `sudo nmap -O 172.16.4.242`

Explanation: Analyzes TCP/IP stack characteristics to guess the operating system and device type.

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -O 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:45 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00057s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
5357/tcp open  wsdapi
7070/tcp open  realserver
MAC Address: 70:B5:E8:58:22:8E (Dell)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2019|10|XP (91%)
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_xp::sp3
Aggressive OS guesses: Microsoft Windows Server 2019 (91%), Microsoft Windows 10 1909 (90%), Microsoft Windows XP SP3 (85%), Microsoft
Windows XP SP2 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.53 seconds
```

Figure 9: Enter Caption

Observation: The machine at 172.16.4.242 is a Dell system running a Windows OS, very likely Windows 10 or Server 2019, based on available services and fingerprints.

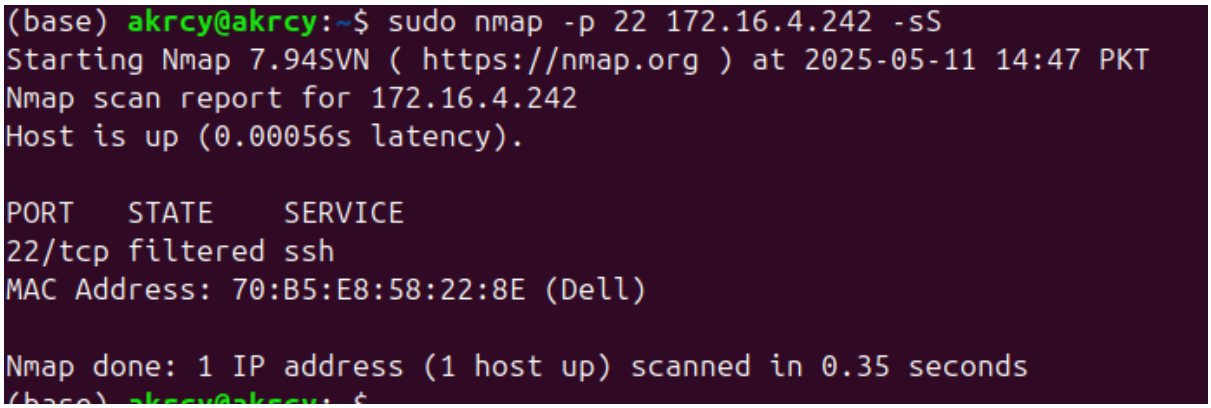
5 Advanced Scanning Techniques

5.1 SYN Scan

Command: `sudo nmap -p 22 172.16.4.2422 -sS`

Explanation: A stealth scan that sends SYN packets without completing the TCP handshake, reducing detectability.

Screenshot:



```
(base) akrcy@akrcy:~$ sudo nmap -p 22 172.16.4.242 -sS
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 14:47 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00056s latency).

PORT   STATE    SERVICE
22/tcp filtered ssh
MAC Address: 70:B5:E8:58:22:8E (Dell)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

Figure 10: Enter Caption

Observation: Port 22 (SSH) is not accessible — most likely blocked by a firewall or access control list (ACL). The system does not appear to be running an SSH server, or access to it is tightly restricted.

6 Packet Analysis with Wireshark

Task: Capture and analyze packets during an Nmap SYN scan (`sudo nmap -p 22 172.16.4.2422 -sS`).

Explanation: Wireshark captures network traffic to verify scan behavior, such as SYN and SYN-ACK packets for open ports.

Screenshot:

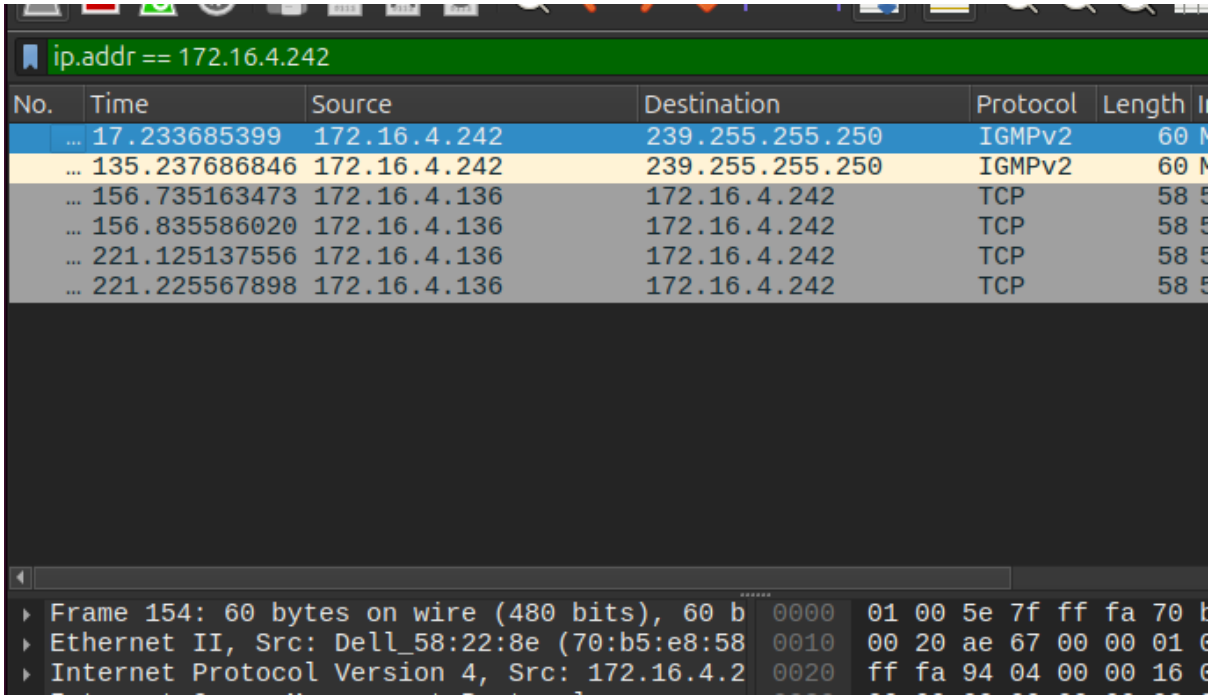


Figure 11: Enter Caption

Observation: The capture showed SYN packets sent to port 22 and SYN-ACK responses, confirming the port was open.

7 Reporting

7.1 Text Report

Command: `sudo nmap -oN samplereport.txt 172.16.4.242`

Explanation: Saves scan results in a text file for documentation and sharing.

Screenshot:

Observation: The text file was easy to read and included all relevant scan details.

```
(base) akrcy@akrcy:~$ sudo nmap -oN samplereport.txt 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 15:02 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00045s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
5357/tcp open  wsdapi
7070/tcp open  realserver
MAC Address: 70:B5:E8:58:22:8E (Dell)

Nmap done: 1 IP address (1 host up) scanned in 4.88 seconds
(base) akrcy@akrcy:~$ ls samplereport.txt
samplereport.txt
(base) akrcy@akrcy:~$
```

Figure 12: Enter Caption

7.2 XML Report

Command: `sudo nmap -oX report.xml 172.16.4.242`

Explanation: Saves results in XML format for integration with other tools.

Screenshot:

```
(base) akrcy@akrcy:~$ sudo nmap -oX samplereport.xml 172.16.4.242
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-05-11 15:03 PKT
Nmap scan report for 172.16.4.242
Host is up (0.00036s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
5357/tcp open  wsdapi
7070/tcp open  realserver
MAC Address: 70:B5:E8:58:22:8E (Dell)

Nmap done: 1 IP address (1 host up) scanned in 4.66 seconds
(base) akrcy@akrcy:~$ ls samplereport.xml
samplereport.xml
(base) akrcy@akrcy:~$
```

Figure 13: Enter Caption

Observation: The XML format was structured and suitable for automated processing.